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(54) **AI DRIVEN PAGE NAVIGATION BASED ON USER BEHAVIOR**

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CPC **G06F 16/954** (2019.01); **G06F 16/9574** (2019.01)

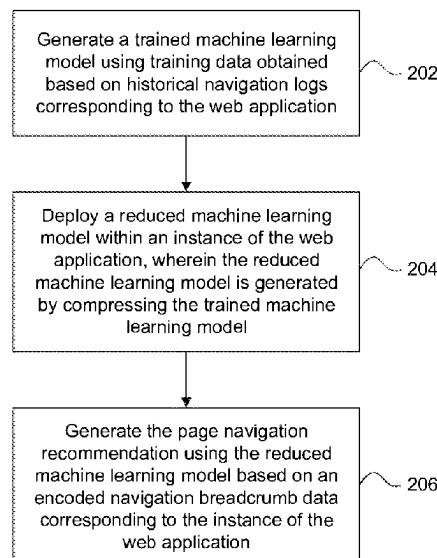
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See application file for complete search history.

(57) **ABSTRACT**

Disclosed herein are system, method, and computer program product embodiments for implementing AI driven application navigation recommendations based on user behavior. An embodiment operates by generating a trained machine learning model using training data obtained based on historical navigation logs corresponding to the web application. The embodiment deploys a reduced machine learning model within an instance of the web application, and the reduced machine learning model is generated by compressing the trained machine learning model. The embodiment then generates the page navigation recommendation using the reduced machine learning model based on an encoded navigation breadcrumb data corresponding to the instance of the web application.

17 Claims, 3 Drawing Sheets

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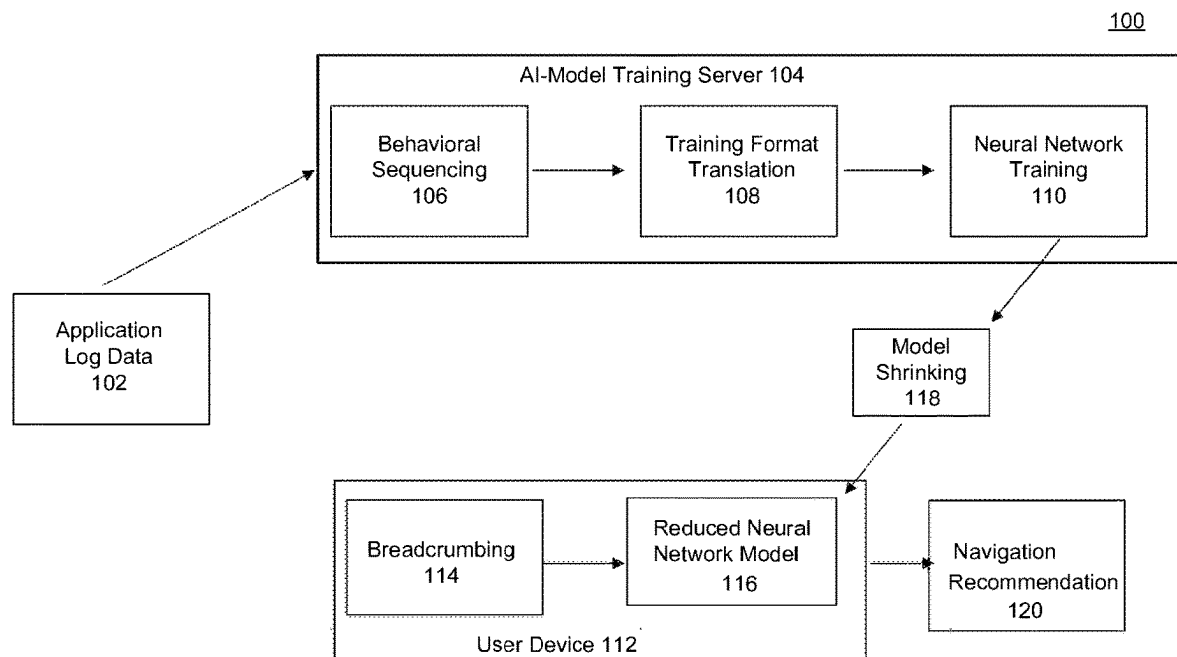
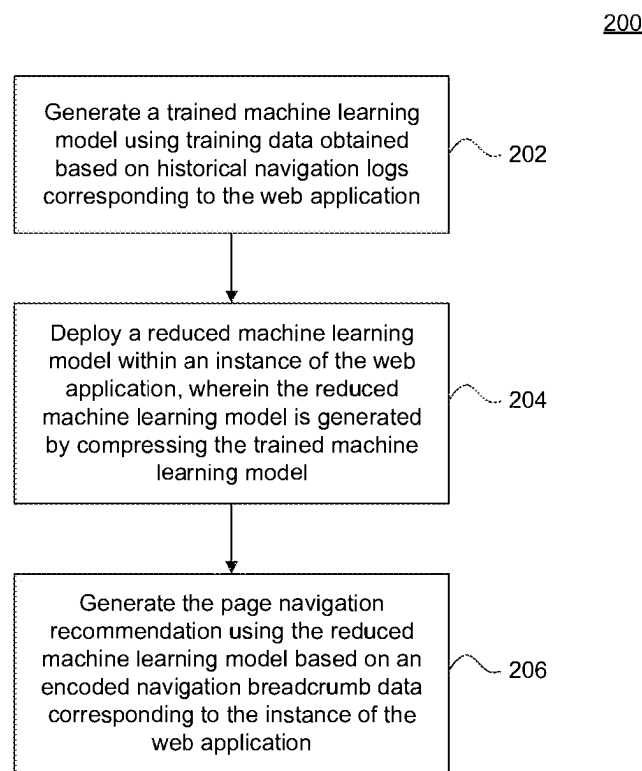
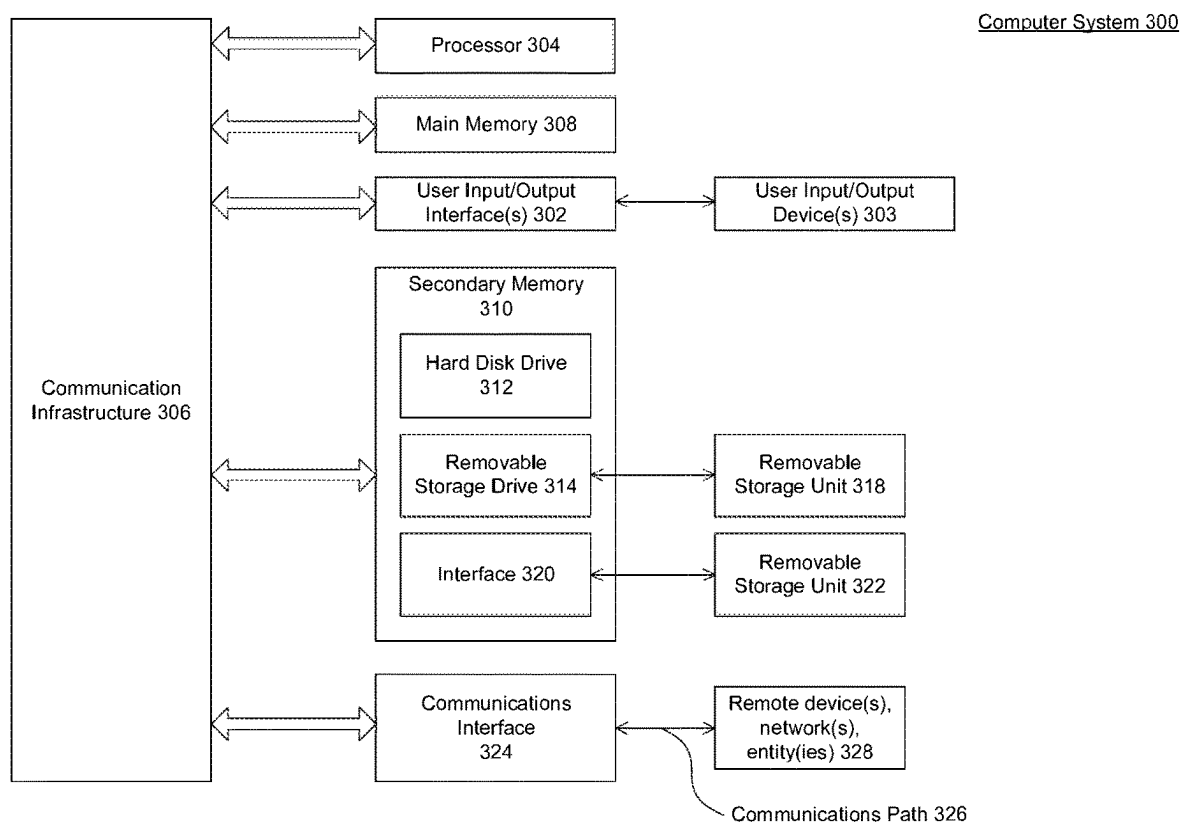


FIG. 1

**FIG. 2**

**FIG. 3**

AI DRIVEN PAGE NAVIGATION BASED ON USER BEHAVIOR

BACKGROUND

One or more implementations relate to the field of enterprise web application design, and more specifically, to page navigation design in enterprise web applications.

Modern enterprise web applications are complex systems with hundreds of pages and a rich set of features. Due to the complexity of an enterprise application, a new user might be faced with an overwhelming number of navigation possibilities. Difficulty in navigating and using an application may hinder its large-scale adoption. Accordingly, a need exists to provide end users with intelligent solutions that would improve ease of use and navigation.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are incorporated herein and form a part of the specification.

FIG. 1 illustrates a block diagram of an exemplary environment **100** for implementing a neural network model to generate a page navigation recommendation, in accordance with some embodiments.

FIG. 2 illustrates a flowchart diagram of an exemplary method for implementing a neural network model to generate a page navigation recommendation, according to some embodiments.

FIG. 3 depicts an example computer system useful for implementing various embodiments.

In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

Provided herein are system, apparatus, device, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for implementing AI driven application navigation recommendations based on user behavior.

Enterprise web applications may contain hundreds of pages and features that a user can navigate to, making their usage potentially extremely complex. Navigating a new interface for the first time may be overwhelming with too many options. However, historical application log data can be leveraged to improve the usability and ease of navigation of an enterprise web application. For example, application log data that documents organization-wide user navigation behavior may be used to train an AI or a neural network model to generate an intelligent page navigation recommendation with an application. In addition, a lightweight reduced version of the trained neural network model may subsequently be ported to a user device to generate page recommendations at the user device.

Various embodiments of these features will now be discussed with respect to the corresponding figures.

FIG. 1 illustrates an example system **100** for implementing a machine learning model to generate a page navigation recommendation, in accordance with some embodiments. The system **100** is merely an example of one suitable system environment and is not intended to suggest any limitation as to the scope of use or functionality of aspects described herein. Neither should the system **100** be interpreted as

having any dependency or requirement related to any single module/component or combination of modules/components described therein.

The system **100** may include an AI-model training server **104** and a client device **112**. According to some aspects, the AI-model training server **112** may include a behavioral sequencing module **106**, a training format translation module **108**, and a neural network training module **110**. The application log data **102** is used as training input by the neural network training module **110** to generate a trained machine learning or a trained neural network model. The user device **112** may import and implement a reduced version of the trained model to generate a navigation recommendation **120**. According to some aspects, the client device **112** may be a mobile phone, a personal computer, a laptop computer, or other types of computing or communication devices. The client device **112** can be a portable electronic device, or a browser located in the portable electronic device.

Application log data **102** documents users' interaction with the enterprise web application. According to some aspects, the application log data **102** may be generated to measure page loading performance and includes information corresponding to application users' navigational behavior. According to some aspects, when a user navigates to page, a page view log may be created and stored on an application log database. According to some aspects, an entry in the application log may be created in response to a user actions (e.g., selecting a tab, checking a box, performing a mouse click event, etc.). According to some aspects, each navigable component of the enterprise application may be assigned a unique identifier. A unique identifier may be associated with each page, tab, feature, and/or the like of the enterprise application (e.g., page ID, tab ID, feature ID, etc.). Each user that interacts with the enterprise application and/or each user-session of the enterprise application may be identified with a unique identifier (e.g., a user ID, a session ID, etc.).

According to some aspects, an application log server may store application log data obtained based on application interaction activity generated by all users within an organization. According to some aspects, each entry of the application log may include an identifier corresponding a component of the application (e.g., a page ID, a tab ID, a feature ID, etc.), an identifier corresponding to a user interacting with the application (e.g., user ID, user-role ID, session ID, etc.), and a time stamp denoting the date and time of the day at which the user interaction/activity has occurred.

According to some aspects, behavioral sequencing module **106** process the application log data to identify users' behavioral patterns as they interact with the enterprise application. Data corresponding to users' historical behavioral patterns may be used to train a machine learning or a neural network model to generate navigation recommendations. Also, the application log data may also include information that is not relevant and/or useful for training a machine learning or a neural network model. According to some aspects, the behavioral sequencing module **106** may perform data mining operations on the application log data to extract user behavior patterns. According to some aspects, the behavioral sequencing module may perform clustering operations to identify user behavior patterns. The behavioral sequencing module may perform clustering operations on the application log data based on user IDs and/or session IDs. The behavioral sequencing module **106** may then identify sequences of user actions (e.g., sequences of page views and/or tabs views, and the like) from the clustered application log entries. According to some aspects, all

application log entries that were generated within a pre-defined interval of time, and associated with a single user ID, may be clustered into a user session log data.

According to some aspects, the behavior sequence module 106 may identify and extract users' behavioral patterns from the clustered application log entries using the time stamp information in the application log data 102. According to some aspects, information corresponding to a user behavioral pattern or a user action sequence may include page IDs corresponding to a chronological sequence of pages of the enterprise application that were visited by the user. For each page in a user action sequence, the behavioral sequence module 106 may also identify information regarding the duration of time the page was kept open by a user. According to some aspects, a user action sequences may be generated from the application log data based on one or more parameters such as the time of the day, user ID, and the role of the user within the organization. According to some aspects, a data mining procedure may be applied to cluster application log data 102 to identify user behavior sequences based on one or more parameters such as the time of the day, user IDs, and/or user-role IDs.

According to some aspects, the training format translation module 108 converts the user action sequences identified by the behavioral sequencing module 106 into numeric vectors or tensors of user actions which are applicable for machine learning training. According to some aspects, the numeric vectors or tensors of user actions may be as used as inputs for training a machine learning or a neural network model using a supervised or unsupervised learning approach. According to some aspects, the numeric vectors of a training data set may include information corresponding to a time of the day and a role of a user, along with numerical information corresponding to user action sequences, such as page IDs, tab IDs, and/or the like. According to some aspects, the training data input may be matrix having the numeric vectors or tensors of user actions as row vectors. According to some aspects, the numeric vectors or tensors may be decimal, hexadecimal, or binary vectors or sequences.

According to some aspects, the training format translation module 108 may generate a numeric vector corresponding to each user action sequence identified by the behavioral sequencing module 106. According to some aspects, each numeric vector may include a predefined number of page IDs and/or tab IDs in a binary or non-binary format. According to some aspects, each numeric vector may include a predefined number of identifiers, each corresponding to a user action in a user action sequence. For example, each numeric vector may be a predefined length sequence of page IDs and/or tab IDs. According to some aspects, the numeric vectors or tensors of user actions may include data corresponding to the time of the day information and/or user-role information. As a non-limiting example, a numeric vector may include the following variables/fields: [user_role_ID, time of the day, ID_1, ID 2, . . . , ID_N, next ID]. According to some aspects, the variable "user_role_ID" may uniquely identify the role of user within the organization, and the variable "time of the day" may identify the hour of the day. The "N" variables, "ID_1" through "ID_N," may correspond to identifiers of the sequence of pages/tabs visited by a user. According to some aspects, each variable/field of a numeric vector may have a predefined length (e.g., all identifiers are 4-bit binary sequences).

According to some aspects, when using the generated numeric vectors as input training data for supervised training of the neural network model, the user-role information, the time of the day information, and the sequence of "N"

page/tab identifiers may be tagged as input elements and the page/tab identifier corresponding to "next-ID" may be tagged as an output for the model. According to some aspects, the vector sequences used as training data do not include information specific to users, such as User IDs and the like. According to some aspects, the training data input may be in the form of matrix with the first "N+2" columns corresponding to the tagged as training inputs and the last column may be tagged as the training output.

According to some aspects, the numeric vectors or tensors of user actions generated by the training format translation module 108 may be used by the neural network training module 110 to train a neural network model. According to some aspects, the training format translation module 108 may use numeric vectors or tensors of user actions of various lengths (e.g., numeric vectors having various "N" values.) to train the neural network model. According to some aspects, the neural network training module 110 generates a trained TensorFlow model using the numeric vectors or tensors of user actions as training data. According to some aspects, the neural network training module 110 may bypass using tensors generated by the training format translation module 108 to perform model training using an unsupervised approach. To perform unsupervised training, the neural network training module 110 may directly use the application log data 102 as input.

According to some aspects, the trained neural network model generated at the AI-model training server 104 is reduced and translated into a compressed/reduced neural network model 116. The compressed/reduced neural network model 116 is a lightweight device level model that is deployed at the user device 112. The reduced neural network model 116 may use significantly less computational power, memory, and/or energy consumption compared to an uncompressed ML model. According to some aspects, the reduced neural network model 116 may be used as part of a web browser at the user device 112.

According to some aspects, the model shrinking module 118 may generate the reduced neural network model 116 using a pruning, quantization, knowledge distillation, and/or factorization based compression approach. According to some aspects, pruning based model compression involve removing certain connections from a trained network or the neural network model. A quantization based model compression approach involves decreasing the size of the weights corresponding to the trained model or network. A distillation based model compression involves transferring knowledge from a large trained model to a smaller model for deployment by training the smaller model to emulating the outputs of the larger model. A factorization based compression approach may use a low-rank approximation to reduce parameter count and simplify the structure of the trained model.

According to some aspects, the reduced neural network model 116 may be a TensorFlow Lite model. A trained TensorFlow model is converted into a TensorFlow Lite model using a pruning, quantization, knowledge distillation, and/or factorization based compression approach.

According to some aspects, navigation breadcrumb information generated at the client device 112 is used as input to the reduced neural network model 116 to generate the navigation recommendation 120. According to some aspects, a breadcrumb trail is a record of a user's navigation history. A breadcrumb trail may be a linear link identification consisting of a row of page IDs and/or tag IDs. According

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to some aspects, the link identification information represents some aspects of a user's current state within an application.

According to some aspects, once a compressed model is deployed at the end device **112**, an on-device breadcrumb log is defined and stored. According to some aspects, a breadcrumb log generated at the end device **112** may be similar to the application log data generated at the application log server **104**, but contains only the necessary information for the compressed model (e.g., a user action sequence of a predefined length.). According to some aspects, at runtime, as the breadcrumb logs are generated, they are encoded and translated into a format that is consumable for the compressed model deployed in software at the end device **112**. The reduced breadcrumb log contains only the necessary information which is needed for the model and is encoded and formatted into a breadcrumb tensor, which is the same data format used to train the model.

According to some aspects, the breadcrumb module **114** generates a breadcrumb log based on the user's interaction with an instance of the enterprise web application implemented at the user device **112**. According to some aspects, the breadcrumb information is translated into a format that is similar to the numeric vectors or tensors of user actions generated by the training format translation module **108**. For example, each breadcrumb numeric vector or tensor may be a predefined length sequence of page IDs and/or tab IDs. According to some aspects, the input to the reduced neural network model **116** may include information corresponding to the time of the day and/or the role of the user interacting with the instance of the web application at the user device **112**. As a non-limiting example, inputs to the reduced neural network model **116** may include the following variables/fields: [user_role_ID, time of the day, ID_1, ID_2, . . . , ID_N]. The variable "user_role_ID" may identify the role of user that is interacting with the instance of the enterprise web application at the user device **112**, and the variable "time of the day" may identify the time of the interaction. The "N" variables, "ID_1" through "ID_N," may correspond to identifiers of the sequence of pages/tabs visited by a user at the user device **112**. According to some aspects, the AI-model training server **104** provides the user device **112** with a recommended input numeric vector length (e.g., value of N). According to some aspects, the length of the input vectors may be increased to improve accuracy of the generated recommendations.

According to some aspects, a request to generate a navigation recommendation **120** may be sent to the reduced neural network model **116** from within the instance of the enterprise web application that is running at the user device **112**. According to some aspects, the navigation recommendation **120** may include a list of potential pages and/or tabs that the user may navigate to. According to some aspects, the listed pages and/or table in the navigation recommendation **120** may also include an associated probability and/or a confidence value that indicates the confidence level the model associates with the generated recommendation. According to some aspects, a user may provide feedback regarding their perceived usefulness/accuracy of the navigation recommendation **120**. According to some aspects, the reduced neural network model **116** may use a set of input numeric vectors and the navigation recommendations that were generated in response to the input vectors to train and/or fine-tune the reduced neural network model locally at the user device **112**. According to some aspects, user feedback may be used to train and/or fine-tune the reduced neural network model locally at the user device **112**.

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FIG. **2** is a flowchart for method **200** for implementing API self-paced migration, according to an embodiment. Method **200** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. **2**, as will be understood by a person of ordinary skill in the art.

Method **200** shall be described with reference to FIG. **1**. However, method **200** is not limited to those example embodiments.

In step **202**, a trained machine learning or a trained neural network model is generated by the neural network training module **110** using training data obtained based on historical navigation logs corresponding to the web application. According to some aspects, the historical navigation logs are the application log data **102** that are created in response to application interaction activity generated by all users within an organization.

According to some aspects, each entry of the application log may include a page ID or a tab ID, a user ID, and a time stamp denoting the date and time of the day at which the user interaction/activity has occurred. According to some aspects, the behavior sequence module **106** may use the time stamp information in the application log data **102** to cluster the application logs and/or identify sequences of user actions. According to some aspects, information corresponding to a user action sequence may include page IDs corresponding to a chronological sequence of pages of the enterprise application that were visited by the user.

According to some aspects, the training format translation module **108** converts the user action sequences identified by the behavioral sequencing module **106** into numeric vectors or tensors of user actions. Each numeric vector generated by the training format translation module **108** may correspond to a sequence of pages of the enterprise web application visited by the user. The numeric vectors or tensors of user actions generated by the training format translation module **108** may be used as training data by the neural network training module **110** to generate the a trained machine learning or a trained neural network model.

In step **204**, a reduced machine learning or neural network model is deployed within an instance of the web application deployed at user device **112**. According to some aspects, the reduced machine learning model is generated by the model shrinking module **118** using the trained machine learning model generated by the neural network training module **110**. According to some aspects, the reduced machine learning model is generated using one or more of a quantization based algorithm, a factorization based algorithm, a distillation based algorithm, or a pruning based algorithm. According to some aspects, the reduced neural network model **116** may be a TensorFlow Lite model.

In step **206**, the navigation recommendation **120** may be generated by the reduced machine learning model using encoded navigation breadcrumb data generated at the user device **112**. According to some aspects, the encoded navigation breadcrumb data is generated from breadcrumb logs that are created based on the user's interaction with an instance of the enterprise web application implemented at the user device **112**. According to some aspects, the breadcrumb information is translated into a format that is similar to the numeric vectors or tensors of user actions generated by the training format translation module **108**. For

example, each breadcrumb numeric vector or tensor may be a predefined length sequence of page IDs and/or tab IDs. According to some aspects, the input to the reduced neural network model **116** may include information corresponding to the time of the day and/or the role of the user interacting with the instance of the web application at the user device **112**.

According to some aspects, a navigation recommendation **120** includes multiple pages and/or tabs. Each of the pages and/or tabs includes in the navigation recommendation may be associated with a confidence value generated by the reduced neural network model. The user may select, for the navigation recommendation **120**, a page or a tab to navigate to next based on an associated confidence value. According to some aspects, to improve the relevance and accuracy of the navigation recommendations **120**, the reduced machine learning model **120** may be further fine-tuned based on the input navigation breadcrumb data and the generated navigation recommendations **120**.

Various embodiments may be implemented, for example, using one or more well-known computer systems, such as computer system **300** shown in FIG. 3. One or more computer systems **300** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

Computer system **300** may include one or more processors (also called central processing units, or CPUs), such as a processor **304**. Processor **304** may be connected to a communication infrastructure or bus **306**.

Computer system **300** may also include user input/output device(s) **303**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **306** through user input/output interface(s) **302**.

One or more of processors **304** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

Computer system **300** may also include a main or primary memory **308**, such as random access memory (RAM). Main memory **308** may include one or more levels of cache. Main memory **308** may have stored therein control logic (i.e., computer software) and/or data.

Computer system **300** may also include one or more secondary storage devices or memory **310**. Secondary memory **310** may include, for example, a hard disk drive **312** and/or a removable storage device or drive **314**. Removable storage drive **314** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

Removable storage drive **314** may interact with a removable storage unit **318**. Removable storage unit **318** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **318** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **314** may read from and/or write to removable storage unit **318**.

Secondary memory **310** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **300**. Such

means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **322** and an interface **320**. Examples of the removable storage unit **322** and the interface **320** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

Computer system **300** may further include a communication or network interface **324**. Communication interface **324** may enable computer system **300** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **328**). For example, communication interface **324** may allow computer system **300** to communicate with external or remote devices **328** over communications path **326**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **300** via communication path **326**.

Computer system **300** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

Computer system **300** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software (“on-premise” cloud-based solutions); “as a service” models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

Any applicable data structures, file formats, and schemas in computer system **300** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **300**, main memory **308**, secondary memory **310**, and removable storage units **318** and **322**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **300**), may cause such data processing devices to operate as described herein.

Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. 3. In particular, 5 embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more 10 but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

While this disclosure describes exemplary embodiments 15 for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, 20 embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein. 25

Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alter- 30 nate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein. 35

References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, 40 or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such 45 feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as 50 synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements 55 are not in direct contact with each other, but yet still co-operate or interact with each other.

The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the 60 following claims and their equivalents.

What is claimed is:

1. A method for generating a page navigation recommendation for a web application at a user device, comprising: 65
deploying, at the user device, a reduced machine learning model within an instance of the web application, wherein the reduced machine learning model is gener-

ated by compressing a trained machine learning model generated using training data obtained based on historical navigation logs corresponding to the web application;

encoding, at the user device, a navigation breadcrumb log corresponding to the instance of the web application to generate an encoded navigation breadcrumb log consumable for the reduced machine learning model, wherein the encoded navigation breadcrumb log comprises a same data format as the historical navigation logs;

generating, at the user device, the page navigation recommendation and a confidence value associated with the page navigation recommendation by applying the encoded navigation breadcrumb log to the reduced machine learning model;

tuning, at the user device based on the encoded navigation breadcrumb log, the page navigation recommendation, and a user feedback, the reduced machine learning model to increase the confidence value associated with the page navigation recommendation, the user feedback comprising selecting, based on the confidence value, a next page to which the page navigation recommendation should navigate;

generating, at the user device, another page navigation recommendation using the tuned reduced machine learning model based on another encoded navigation breadcrumb log corresponding to the instance of the web application; and

outputting, at the user device, the other page navigation recommendation.

2. The method of claim 1, wherein the generating the page navigation recommendation further comprises:

determining, by the reduced machine learning model, a plurality of next-page predictions and a plurality of corresponding confidence values; and

selecting as the page navigation recommendation a subset of the plurality of next-page predictions based on the plurality of corresponding confidence values.

3. The method of claim 1, further comprising:

training, at the user device, the reduced machine learning model based on the encoded navigation breadcrumb log.

4. The method of claim 1, wherein the reduced machine learning model is generated by compressing the trained machine learning model using one or more of a quantization based algorithm, a factorization based algorithm, a distillation based algorithm, or a pruning based algorithm.

5. The method of claim 1, wherein the encoded navigation breadcrumb log comprises a numeric vector corresponding to a sequence of pages visited at the user device.

6. The method of claim 1, wherein an entry of the historical navigation logs comprises a user identifier, a page identifier, and a time stamp.

7. The method of claim 1, wherein the trained machine learning model is generated based on a neural network modeling.

8. A system for generating a page navigation recommendation for a web application at a user device, comprising: a memory; and

at least one processor coupled to the memory and configured to:

deploy, at the user device, a reduced machine learning model within an instance of the web application, wherein the reduced machine learning model is generated by compressing a trained machine learn-

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ing model generated using training data obtained based on historical navigation logs corresponding to the web application;

encode, at the user device, a navigation breadcrumb log corresponding to the instance of the web application to generate an encoded navigation breadcrumb log consumable for the reduced machine learning model, wherein the encoded navigation breadcrumb log comprises a same data format as the historical navigation logs;

generate, at the user device, the page navigation recommendation and a confidence value associated with the page navigation recommendation by applying the encoded navigation breadcrumb log to the reduced machine learning model;

tune, at the user device based on the encoded navigation breadcrumb log, the page navigation recommendation, and a user feedback, the reduced machine learning model to increase the confidence value associated with the page navigation recommendation, the user feedback comprising selecting, based on the confidence value, a next page to which the page navigation recommendation should navigate;

generate, at the user device, another page navigation recommendation using the tuned reduced machine learning model based on another encoded navigation breadcrumb log corresponding to the instance of the web application; and

output, at the user device, the other page navigation recommendation.

9. The system of claim 8, wherein to generate the page navigation recommendation, the at least one processor is further configured to;

determine a plurality of next-page predictions and a plurality of corresponding confidence values; and

select as the page navigation recommendation a subset of the plurality of next-page predictions based on the plurality of corresponding confidence values.

10. The system of claim 8, wherein the at least one processor is further configured to:

train, at the user device, the reduced machine learning model based on the encoded navigation breadcrumb log.

11. The system of claim 8, wherein the reduced machine learning model is generated by compressing the trained machine learning model using one or more of a quantization based algorithm, a factorization based algorithm, a distillation based algorithm, or a pruning based algorithm.

12. The system of claim 8, wherein the encoded navigation breadcrumb log comprises a numeric vector corresponding to a sequence of pages visited at the user device.

13. The system of claim 8, wherein an entry of the historical navigation logs comprises a user identifier, a page identifier, and a time stamp.

14. A non-transitory computer-readable medium (CRM) having instructions stored thereon that, when executed by at

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least one computing device, causes the at least one computing device to perform operations comprising:

deploying, at a user device, a reduced machine learning model within an instance of a web application running, wherein the reduced machine learning model is generated by compressing a trained machine learning model generated using training data obtained based on historical navigation logs corresponding to the web application;

encoding, at the user device, a navigation breadcrumb log corresponding to the instance of the web application to generate an encoded navigation breadcrumb log consumable for the reduced machine learning model, wherein the encoded navigation breadcrumb log comprises a same data format as the historical navigation logs;

generating, at the user device, the page navigation recommendation and a confidence value associated with the page navigation recommendation by applying the encoded navigation breadcrumb log to the reduced machine learning model;

tuning, at the user device based on the encoded navigation breadcrumb log, the page navigation recommendation, and a user feedback, the reduced machine learning model to increase the confidence value associated with the page navigation recommendation, the user feedback comprising selecting, based on the confidence value, a next page to which the page navigation recommendation should navigate;

generating, at the user device, another page navigation recommendation using the tuned reduced machine learning model based on another encoded navigation breadcrumb log corresponding to the instance of the web application; and

outputting, at the user device, the other page navigation recommendation.

15. The non-transitory CRM of claim 14, wherein the generating the page navigation recommendation further comprises:

determining, by the reduced machine learning model, a plurality of next-page predictions and a plurality of corresponding confidence values; and

selecting as the page navigation recommendation a subset of the plurality of next-page predictions based on the plurality of corresponding confidence values.

16. The non-transitory CRM of claim 14, wherein the reduced machine learning model is generated by compressing the trained machine learning model using one or more of a quantization based algorithm, a factorization based algorithm, a distillation based algorithm, or a pruning based algorithm.

17. The non-transitory CRM of claim 14, wherein the encoded navigation breadcrumb log comprises a numeric vector corresponding to a sequence of pages visited at the user device.

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